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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Machine Learning (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

Week 9: Assignment 9 (Non Graded)

Assignment not submitted

Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) Consider the bayesian network shown below.

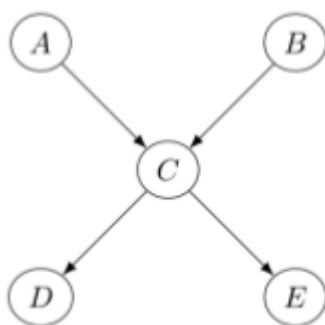
1 point

Figure 1: Q1

Two students - Student A and Student B make the following claims:

- Student A claims $P(A|\{B, D\}) = P(A|D)$
- Student B claims $P(A|B) = P(A)$

where $P(X|Y)$ denotes probability of event X given Y . Please note that Y can be a set. Which of the following is true?

- ☐ Student A and Student B are correct.
- ☐ Student A is correct and Student B is incorrect.
- ☐ Student A is incorrect and Student B is correct.
- ☐ Both are incorrect.

☐ Undirected Graphical Models - Introduction and Factorization (unit? unit=104&lesson=105)

☐ Undirected Graphical Models - Potential Functions (unit? unit=104&lesson=106)

☐ Hidden Markov Models (unit? unit=104&lesson=107)

☐ Variable Elimination (unit? unit=104&lesson=108)

☐ Tree Width and Belief Propagation (unit? unit=104&lesson=109)

☐ **Practice:**
Week 9:
Assignment 9 (Non Graded)
(assessment? name=272)

☐ Week 9 Feedback Form : Introduction To Machine Learning (unit? unit=104&lesson=290)

☒ Week 9: Solution (unit? unit=104&lesson=228)

☐ Insufficient information to make any conclusion. Probability distributions of each variable should be given.

2) Consider the Bayesian graph shown below in Figure 2.

1 point

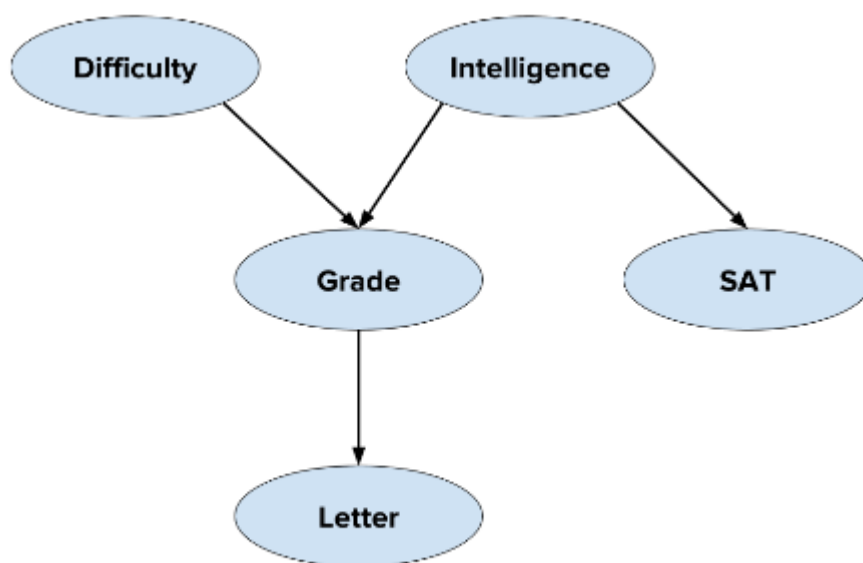


Figure 2: Q3

The random variables are modeled as discrete variables and the corresponding CPDs are as below.

d^0	d^1
0.6	0.4

i^0	i^1
0.6	0.4

	g^1	g^2	g^3
i^0, d^0	0.3	0.4	0.3
i^0, d^1	0.05	0.25	0.7
i^1, d^0	0.9	0.08	0.02
i^1, d^1	0.5	0.3	0.2

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

	l^0	l^1
g^1	0.2	0.8
g^2	0.4	0.6
g^3	0.99	0.01

What is the probability of $P(i = 1, d = 0, g = 2, s = 1, l = 0)$?

☐ 0.004608

☐ 0.006144

Quiz: Week 9 :
Assignment 9
(assessment?
name=301)

Week 10 ()

Week 11 ()

Week 12 ()

Text
Transcripts
()

Download
Videos ()

Books ()

Problem
Solving
Session -
July 2024 ()

- ☐ 0.003992
☐ 0.007309
☐ None of these

3) Consider the Markov network shown below in Figure 3

1 point

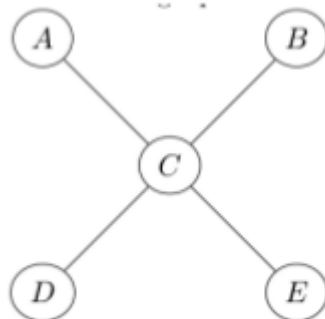


Figure 3: Q4

Two students - Student A and Student B make the following claims:

- Student A claims $P(A|B) = P(A)$
- Student B claims $P(A|\{B, C\}) = P(A|C)$

Which of the following is true?

- ☐ Student A and Student B are correct.
☐ Student A is incorrect and Student B is correct.
☐ Student A is correct and Student B is incorrect.
☐ Both are incorrect.
☐ Insufficient information to make any conclusion. Probability distributions of each variable should be given.

4) In the Markov network given in Figure 4, two students make the following claims: 1 point

- Manish claims variable "1" is independent of variable "6" given variable "3".
- Trina claims variable "1" is independent of variable "8" given variable "4".

Which of the following is true?

- ☐ Both the students are correct.
☐ Trina is incorrect and Manish is correct.
☐ Trina is correct and Manish is incorrect.
☐ Both the students are incorrect.
☐ Insufficient information to make any conclusion. Probability distributions of each variable should be given.

5) Does there exist a more compact factorization involving less number of factors for the distribution given in Equation 1 in previous question? 1 point

- ☐ Yes
☐ No

☐ Insufficient information

Check Answers and Submit

